

1. Bright Image - Grey level Transformation

Answer:

The most suitable gray level transformation is the Negative Transformation (Image negative)

A negative transformation reverses the gray levels of an image by converting bright pixels into dark pixels and dark pixels into bright pixels.

Formula:

$$S = L - 1 - r$$

r = Input pixel intensity

S = Output pixel intensity

L = Number of gray levels (256 for an 8-bit image)

Application:

1. Read the image
2. Apply the transformation $S = 255 - r$ to every pixel
3. Generate the transformed image.

Expected Outcomes:

- * Excessively bright regions become darker.
- * Hidden details in bright areas become visible.
- * Overall visibility and contrast improve.

Negative transformation is useful when important details are present in bright regions, such as X-ray images, satellite images, and scanned documents.

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2. Dark image - Grey level Transformation

The appropriate grey level transformation is the log Transformation. log Transformation expands dark pixel values while compressing brighter ones, making hidden details in low intensity regions visible.

Formula:

$$S = c \log(1+r)$$

- r = Input intensity
- S = Output intensity
- c = Scaling constant

Application:

1. Read the dark image
2. Apply the logarithmic transformation to every pixel
3. Normalize the output intensity values.

It improves low-Intensity Regions:

- * low-gray-level pixels are stretched
- * Small intensity differences become larger.
- * Dark areas reveal more details without excessively brightening the image.

Outcome:

- * Better visibility in shadow regions
- * Enhanced fine details.
- * Improved overall brightness

Log transformation is ideal for enhancing dark images, medical images, astronomical images, and images captured under poor lighting conditions.

3. Poor Contrast - Histogram Technique

The suitable histogram-based technique is histogram equalization. Histogram Equalization redistributes pixel intensity value so that the histogram becomes more uniform, thereby improving image contrast.

Working process.

1. Compute the histogram of the image
2. Calculate the cumulative distribution function (CDF).
3. Map old intensity values to new intensity values using the CDF.
4. Generate the enhanced image.

Improves Image Quality:

- * Increases the dynamic image range of intensity values.
- * Makes dark regions brighter and bright regions more distinguishable.
- * Enhances object visibility.

Outcomes:

- * Improved Contrast
- * Better edge visibility
- * More uniform intensity distribution.

Histogram Equalization is widely used in medical imaging, satellite imaging, and low-contrast photographs because it automatically enhances contrast without manual adjustment.

4. low contrast - uniform Intensity.

The appropriate method is Histogram Equalization.

Histogram Equalization transforms the intensity distribution of an image into a nearly uniform distribution, improving overall contrast.

Working process:

1. calculate the image histogram
2. compute the cumulative histogram (CDF).
3. use the CDF to remap pixel intensity values
4. generate the enhanced image.

Effect on intensity levels:

- * Frequently occurring intensity values are spread over a wider range.
- * Pixel values become more uniformly distributed
- * Contrast between neighboring regions increases.

Outcomes:

- * uniform intensity distribution
- * Improved visual appearance
- * Enhanced feature detection.

Histogram Equalization is preferred because it utilizes the full gray-level range and enhances the visibility of details throughout the image.

5. Uneven Brightness - Enhancement Technique

The appropriate enhancement technique is Adaptive Histogram Equalization (AHE) or Contrast Limited Adaptive Histogram Equalization (CLAHE).

Working process:

1. Divide the image into small tiles
2. Perform histogram equalization on each tile independently.
3. Apply a clip limit (CLAHE) to prevent over-enhancement.
4. Merge neighboring tiles smoothly using interpolation

Application:

- * Corrects uneven lighting across different regions.
- * Enhances local contrast
- * Preserves image details while minimizing noise.

Outcome:

- * Uniform brightness across the image
- * Better local contrast
- * Improved visibility in both dark and bright regions.
- * Reduced noise amplification

~~CLAHE~~ CLAHE is preferred over global histogram equalization for images with non-uniform illumination because it enhances local details while preventing excessive contrast enhancement and noise. It is widely used in medical imaging, surveillance, and low-light photography.